

NASHUA, NH, USA

WMO: 743946

Lat: 42.783N

Lon: 71.517W

Elev: 200

StdP: 14.59

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 54754

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.5	(c) 7.2	(d) -11.2	(e) 3.0	(f) 8.5	(g) -7.5	(h) 3.7	(i) 11.8	(j) 25.0	(k) 24.9	(l) 22.5	(m) 26.6	(n) 5.7	(o) 310	(p) 0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	91.8	73.5	88.9	72.0	86.0	70.3	76.5	86.2	74.9	83.8	73.3	81.5	7.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	126.9	80.2	72.4	120.7	78.8	70.8	114.4	77.6	39.9	86.6	38.3	83.9	36.9	81.5	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	18.0	15.6	-6.0	96.8	6.3	2.5	-10.5	98.6	-14.2	100.1	-17.7	101.5	-22.3	103.3
			-6.9	79.2	5.8	1.8	-11.1	80.4	-14.4	81.4	-17.7	82.4	-21.9	83.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	26.0	27.2	35.8	47.9	59.3	67.3	73.9	71.3	63.7	52.1	40.4	31.0
	DBStd	18.59	10.13	9.23	9.43	8.63	8.10	6.75	5.29	5.37	7.47	7.75	8.44	9.13
	HDD50	2953	745	639	453	140	10	0	0	0	0	67	307	592
	HDD65	6281	1209	1057	906	518	217	50	4	9	113	405	738	1055
	CDD50	2878	1	2	12	77	300	520	742	660	413	131	18	2
	CDD65	731	0	0	0	5	42	120	281	204	75	4	0	0
	CDH74	7256	0	0	13	99	556	1212	2782	1868	676	50	0	0
	CDH80	2578	0	0	2	32	189	409	1105	624	210	7	0	0
	WSAvg	5.2	5.7	6.3	6.8	6.5	5.4	4.8	4.2	3.9	4.0	4.5	5.0	5.1
	PrecAvg	46.3	3.4	3.4	3.9	4.1	4.1	3.9	3.6	3.6	3.8	4.6	4.1	4.2
(15) Precipitation	PrecMax	58.8	6.0	11.5	7.2	9.2	11.7	10.9	6.0	9.1	9.1	14.3	7.1	7.0
	PrecMin	32.8	1.0	1.7	0.7	0.6	0.8	0.9	1.5	0.2	1.3	0.4	1.3	1.3
	PrecStd	7.0	1.4	2.0	1.6	2.4	2.4	2.7	1.3	2.3	2.2	3.4	1.5	1.5
	PrecStd	7.0	1.4	2.0	1.6	2.4	2.4	2.7	1.3	2.3	2.2	3.4	1.5	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.6	61.3	73.3	84.7	91.1	94.1	95.7	93.4	91.3	80.1	68.9	61.6
		MCWB	55.6	51.6	55.5	61.6	68.4	74.0	76.1	75.3	74.1	66.1	59.7	57.1
	2%	DB	51.6	51.3	63.0	75.3	85.6	88.8	91.9	89.6	85.9	74.0	63.9	53.6
		MCWB	45.9	42.5	51.9	57.8	65.7	71.4	74.0	72.8	71.1	64.7	56.0	49.1
	5%	DB	44.6	45.1	55.6	69.5	80.2	84.5	89.3	86.2	81.6	69.7	59.6	48.6
		MCWB	39.7	38.6	44.8	54.7	63.7	68.3	72.7	71.3	69.4	63.1	52.7	43.5
	10%	DB	39.7	40.2	50.3	64.5	75.4	81.1	86.5	83.4	77.5	65.1	55.1	44.2
		MCWB	34.9	34.9	42.2	52.3	61.4	66.2	71.2	69.3	66.8	57.9	49.4	40.3
	0.4%	WB	56.8	55.5	59.5	65.9	71.8	76.1	79.4	78.7	76.5	70.8	63.5	57.4
		MCDB	58.0	59.4	70.4	75.4	86.7	88.9	89.6	90.3	85.1	74.7	67.3	59.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.3	44.5	53.5	61.7	69.1	73.9	76.9	76.1	73.8	67.5	58.3	50.6
		MCDB	49.9	50.3	60.5	71.8	79.7	84.8	87.1	84.4	82.1	70.7	61.6	52.9
	5%	WB	40.0	39.3	47.1	57.4	66.7	71.8	75.2	74.3	71.9	63.5	54.3	44.8
		MCDB	44.1	44.1	53.1	65.7	75.9	81.0	84.6	81.3	78.6	68.3	57.8	47.3
	10%	WB	35.6	35.9	42.7	52.9	63.7	69.5	73.6	72.3	69.4	60.1	50.3	40.4
		MCDB	39.0	39.3	49.2	62.0	72.1	75.7	82.2	79.5	74.8	64.1	54.9	43.6
(35) Mean Daily Temperature Range	5% DB	MDBR	17.2	18.0	18.7	21.3	21.8	20.4	20.9	21.1	21.2	19.9	18.2	15.9
		MCDBR	21.6	25.2	27.2	30.5	29.7	25.3	24.3	24.0	25.2	24.5	25.1	21.6
		MCWBR	18.2	18.4	17.0	15.8	13.6	11.0	9.8	10.2	12.6	15.3	18.1	18.1
	5% WB	MCDBR	21.2	23.1	24.1	26.3	25.4	21.9	20.7	19.7	20.7	20.7	20.2	20.2
		MCWBR	18.7	17.7	17.2	16.4	13.3	11.1	9.6	9.8	11.5	15.7	16.9	18.5
(40) Clear-Sky Solar Irradiance	taub		0.284	0.295	0.322	0.363	0.399	0.439	0.460	0.427	0.373	0.332	0.307	0.287
	taud		2.458	2.438	2.451	2.388	2.326	2.241	2.192	2.266	2.398	2.528	2.553	2.541
	Ebn at Noon		274	289	293	287	278	266	258	264	272	272	264	264
	Edh at Noon		23	28	31	36	40	43	45	41	33	26	22	20
	RadAvg		563	836	1158	1464	1612	1745	1843	1662	1344	870	600	456
(45) All-Sky Solar Radiation	RadStd		35	65	77	158	194	169	115	85	92	81	52	48

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (78 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+65

Nomenclature: See separate page